



NeuroCore: A Four-Primitive Deterministic Cognitive Flow Organism for Attention, Clarity, Emotional Load, and Decision Equilibrium

*Attention Flow · Intention Flow · Emotional Flow · Cognitive Load Flow —
The Lume 4/42 Architecture Applied to the Cognitive Domain*

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Companion Papers: BioCore · Meridian Architecture · Meridian as Synthetic Organism · Verdara
Ultra · HydroCore

Related: Lume [10.5281/zenodo.19382282](https://zenodo.org/record/19382282) · Trust Layer [10.5281/zenodo.19560674](https://zenodo.org/record/19560674) · DAIGS
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Patent Pending — DarkWave Studios LLC
Systems and Methods for Deterministic Multi-Layer Wireless Energy Routing
Lume-X: Provisionally patented deterministic control runtime
Patent Pending — DarkWave Studios LLC

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Abstract

I present NeuroCore, a Lume-native deterministic cognitive flow organism for attention, clarity, emotional load, and decision equilibrium governance. NeuroCore implements the Lume 4/42 organism architecture — four irreducible flow primitives governing forty-two operational nodes — in the cognitive domain. The four primitives are Attention Flow, Intention Flow, Emotional Flow, and Cognitive Load Flow. The forty-two nodes encode the complete internal cognitive state space from focus stability and working memory load through goal clarity, decision paralysis risk, emotional reactivity, burnout pressure, and global cognitive resilience.

NeuroCore is the first organism in the Lume-Cognition vertical and occupies the cognitive layer directly above BioCore in the canonical organism stack. Where BioCore governs the biological substrate, NeuroCore governs the cognitive output of that substrate: the attention, intention, emotion, and load that determine whether the mind is clear, stable, and capable of producing coherent decisions. The two organisms are formally coupled: BioCore exports sleep quality, stress load, recovery debt, and inflammation signals to NeuroCore as first-class input nodes, and NeuroCore exports cognitive load and emotional drift back to BioCore's stress/thermal primitive.

Unlike productivity platforms, cognitive performance tools, or mental wellness apps that provide static protocols or reactive readouts, NeuroCore operates as a deterministic organism: it maintains a continuous internal state of cognitive equilibrium across all four primitive flows simultaneously, enforces hard constraint rules that suppress performance-pushing recommendations when any node crosses a critical threshold, selects one of five operating modes deterministically based on node pattern, and evaluates multi-day trends rather than isolated readings before escalating or modifying guidance.

The paper includes a complete formal 2D/3D visual geometry specification: a polar coordinate ring with radial state encoding, four-axis primitive core in a fixed cognitive color palette, primitive envelope volumes, drift vector overlays, and three-dimensional extrusion options for time-stack and depth-stack modes.

Keywords: deterministic cognitive governance, Lume synthetic organism, cognitive flow, attention management, intention alignment, emotional load, decision fatigue,

burnout prevention, 4/42 organism architecture, cognitive equilibrium,
Lume-Cognition vertical, BioCore integration

1 Introduction

1.1 The Problem with Cognitive Governance Today

The contemporary landscape of cognitive performance and mental wellness technology operates under a fundamental mischaracterization of what the mind is and how it fails. Productivity systems treat cognitive capacity as a fixed resource to be allocated across tasks. Wellness apps treat mental state as a scalar readout to be optimized toward a target score. All of these approaches share the same structural flaw: they treat the mind as a static system rather than a dynamic flow.

The mind is not a container with a fixed amount of attention to allocate. It is a continuous field of interacting flows — attention that rises and falls based on sleep, recovery, task switching cost, and sensory interference; intention that drifts when executive function is depleted and goal clarity erodes; emotion that loads and shifts in patterns that compound over days; cognitive pressure that builds from accumulated load and depletes resilience. These flows interact continuously. Depleted attention amplifies decision fatigue. Emotional reactivity destabilizes intention. Burnout pressure collapses the whole system faster than any single depleted node would suggest.

No current cognitive tool maintains a continuous internal model of all four of these flows simultaneously. No current tool enforces hard constraints that protect the cognitive organism from recommendations that would push a depleted system past its safety envelope. NeuroCore is the first Lume organism specification designed to address the cognitive domain with the same architectural rigor applied to energy, water, outdoor, and biological flow governance.

1.2 The Cognitive Flow Reframe

The foundational insight of this work is that cognition is not a performance metric. It is a flow system. Attention is not a score — it is a continuous field that rises and falls with the biological substrate. Intention is not a goal list — it is a flow of executive direction. Emotion is not a mood state — it is a load-bearing flow that stabilizes or destabilizes every other cognitive function. Cognitive load is the pressure under which the entire system operates and the depletion of the resilience envelope.

NeuroCore applies the Lume 4/42 organism architecture to this cognitive flow system. The result is not a productivity tool or a wellness app. It is a cognitive governance organism: a system that maintains equilibrium across all four flows, enforces safety constraints, adapts its outputs deterministically, and never recommends actions that would damage the cognitive organism in order to extract short-term performance.

1.3 Position in the Lume Organism Stack

COGNITIVE LAYER:	NeuroCore (this paper)
BIOLOGICAL LAYER:	BioCore
PHYSICAL LAYER:	Meridian / Verdara Ultra / HydroCore

The physical layer organisms govern the external physical environment. BioCore governs the internal biological substrate. NeuroCore governs the cognitive output of that biological substrate — the mind-layer that emerges from and depends upon the biological layer, and that in turn shapes the biological layer through bidirectional coupling between cognitive and physiological stress.

1.4 Novel Contributions

1. **The first formal 4/42 cognitive flow organism specification** — applying the Lume synthetic organism architecture to the cognitive domain with complete node definitions, threshold bands, and cross-primitive interaction modeling.
2. **Cognitive equilibrium as a governable state** — a formal definition maintained across four flow primitives with a deterministic governance engine.
3. **Hard constraint suppression for cognitive safety** — suppressing performance-pushing guidance when any cognitive safety node crosses a critical threshold.
4. **Five-mode deterministic cognitive operating system** — Focus, Recovery, Emotional Stabilization, Decision, and Cognitive Safety modes selected by node pattern.
5. **Formal BioCore integration specification** — the first organism-to-organism coupling specification in the Lume ecosystem.
6. **Deterministic 2D/3D cognitive geometry** — including polar coordinate ring, four-axis core, drift vector overlays, cognitive pressure visualization, and 3D extrusion modes.

1.5 Scope and Honest Limitations

NeuroCore does not diagnose cognitive or psychiatric disorders. It does not replace clinical neuropsychology, psychiatry, or psychotherapy. The E4 (Depression Load) and E3 (Anxiety Load) nodes are governance signals within the BioCore-NeuroCore wellness framework, not clinical diagnostic instruments. When these nodes cross critical thresholds, NeuroCore explicitly recommends clinical evaluation. All node threshold bands are initial specifications subject to empirical calibration.

2 Background and Related Work

2.1 Attention and Executive Function Research

The cognitive science of attention is among the most thoroughly studied domains in experimental psychology. Kahneman's resource model established the foundational framework of attention as a limited-capacity resource with an allocation policy governed by arousal and effort. Baddeley's working memory model characterized the central executive, phonological loop, and visuospatial sketchpad as the working components of active cognition. Research on sustained attention, mind-wandering, and task-switching cost has produced rich empirical literatures on the conditions under which attentional capacity degrades and recovers.

2.2 Executive Function and Decision-Making

Executive function — the set of cognitive control processes that govern goal-directed behavior, including planning, inhibition, cognitive flexibility, and working memory — is the cognitive substrate of NeuroCore's Intention Flow primitive. Decision fatigue research has documented the progressive depletion of decision quality under sustained decision load.

2.3 Emotional Regulation and Affect Science

Gross's process model of emotion regulation established the foundational taxonomy of regulatory strategies. Research on affective forecasting, emotional contagion, and mood-congruent cognition demonstrates the pervasive influence of emotional state on cognitive performance across all domains. Social baseline theory and evidence for social isolation as a biological and psychological hazard provide grounding for NeuroCore's E7 (Social Overload) and E8 (Social Withdrawal Pressure) nodes.

2.4 Burnout and Cognitive Exhaustion

Maslach's foundational burnout framework — exhaustion, cynicism, and reduced professional efficacy — has been extended by subsequent research to characterize the neurobiological underpinnings of burnout: sustained HPA axis activation, reduced prefrontal function, and structural brain changes following chronic workplace stress. NeuroCore's L3 (Burnout Pressure) node is formally coupled to BioCore's S6 (Burnout Risk) through the BioCore integration model.

2.5 Lume Organism Architecture

The Lume 4/42 organism architecture was established in the energy domain by Meridian and extended to the outdoor, hydrological, and biological domains by Verdara Ultra, HydroCore, and BioCore. BioCore is the immediate predecessor organism in the stack and the primary data source for NeuroCore's biological input nodes.

3 System Overview — The NeuroCore Organism

3.1 Core Thesis

NeuroCore ingests cognitive signals — attention patterns, behavioral indicators of intention coherence, emotional load, and cognitive pressure — maintains a stable internal state of cognitive equilibrium, and outputs deterministic guidance for mental clarity, focus, emotional regulation, decision pacing, and cognitive resilience.

Equilibrium is not peak performance. A practitioner in cognitive equilibrium may be producing far below their technical capability — the organism does not optimize for maximum output. It governs for sustained, safe, resilient function. A governance system that optimizes for output will destroy cognitive resilience over time. NeuroCore optimizes for equilibrium.

3.2 The 4/42 Structure

```
Primitive State Vector N:
  N = [AttentionFlow, IntentionFlow, EmotionalFlow,
        CognitiveLoadFlow]
  Each element is a scalar in [-1.0, +1.0]

Node State Ring R:
  R = [A1..A10, I1..I10, E1..E11, L1..L11]
  Each element is a normalized scalar in [-1.0, +1.0]
```

3.3 Temporal Scale Structure

- **Moment-level organism:** State at the current hour or session. Primary real-time operating resolution.
- **Day-level organism:** Mean or worst-case envelope over the current day. Primary guidance generation resolution.
- **Week-level organism:** Trajectory of day-level organisms over seven days. Trend-detection and pattern-classification resolution.

The cognitive engine's trend-awareness (Rule 6) operates across all three scales, preventing false positive escalation from transient states while ensuring genuine developing patterns are caught early.

4 The Four NeuroCore Primitives

4.1 Attention Flow

Attention Flow governs the quality, stability, and recovery capacity of directed cognitive attention. A positive state indicates stable, sustainable, precise attention. A negative state indicates fragmented, unstable, or fatigued attention. Governed by ten nodes (A1–A10). The most critical are A1 (Focus Stability), A7 (Attention Recovery Speed), and A10 (Attention Fatigue) — the primitive's terminal safety node.

4.2 Intention Flow

Intention Flow governs goal coherence, executive direction, motivation, and the capacity to convert purpose into action. A positive state indicates clear goals, strong motivation, intact executive function, and cognitive confidence to act. A negative state indicates a directionless or fragmented executive system. Governed by ten nodes (I1–I10). The most

safety-critical is I5 (Decision Paralysis Risk). I6 (Internal Narrative Coherence) is among the most important nodes in the full 42-node ring because narrative incoherence is an early warning of multiple downstream failures including burnout, depression load, and intention collapse.

4.3 Emotional Flow

Emotional Flow governs the tone, stability, and regulatory capacity of the affective system. Emotion is not a separate domain from cognition — it is a cognitive primitive. Emotional state shapes memory encoding, decision evaluation, risk weighting, and social interpretation. Governed by eleven nodes (E1–E11). E11 (Emotional Safety Index) is the composite terminal node — when degraded, the practitioner’s cognitive decision space contracts from growth mode to survival mode.

4.4 Cognitive Load Flow

Cognitive Load Flow governs the pressure under which the entire cognitive organism operates and the resilience envelope that determines how much additional load can be safely absorbed. It is the most systemic primitive. Governed by eleven nodes (L1–L11). The most consequential are L3 (Burnout Pressure) — NeuroCore’s primary hard constraint trigger — and L11 (Global Cognitive Resilience), the terminal safety node of the entire ring.

5 The 42-Node Cognitive State Ring

5.1 Attention Flow Nodes (A1–A10)

Node	Name	Description
A1	Focus Stability	Consistency of directed attention on a chosen task. Low stability means involuntary shifts.
A2	Distractibility Index	Susceptibility to interruption by external and internal stimuli.
A3	Task Switching Cost	Cognitive overhead in time, energy, and error rate per context switch.
A4	Working Memory Load	Degree to which working memory buffer is at or near capacity.
A5	Sensory Interference	Volume and intrusiveness of sensory input competing with processing.
A6	Cognitive Drift	Frequency and depth of unprompted mind wandering from task-oriented thinking.
A7	Attention Recovery Speed	Speed of re-establishing directed attention after interruption or drift.
A8	Precision/Detail Orientation	Current capacity for fine-grained, careful cognitive work.
A9	Information Saturation	Point at which additional information reduces rather than improves decision quality.
A10	Attention Fatigue	Cumulative depletion of attentional capacity. Terminal safety node for Attention primitive.

5.2 Intention Flow Nodes (I1–I10)

Node	Name	Description
I1	Goal Clarity	Clear, specific, achievable sense of what the practitioner is working toward.
I2	Motivation Level	Intrinsic and extrinsic motivational energy for goal pursuit.
I3	Executive Function Strength	Capacity of prefrontal executive control: planning, inhibition, flexibility.
I4	Decision Direction	Sense of knowing what to do next — translating goals into immediate actions.
I5	Decision Paralysis Risk	Probability that decision-making has effectively stopped. More options make it worse.
I6	Internal Narrative Coherence	Whether life, work, and relationships form a meaningful, consistent story.
I7	Purpose Alignment	Connection between current activities and deeper values/long-term purpose.
I8	Initiative Momentum	Tendency toward self-initiated action versus passive responsiveness or avoidance.
I9	Cognitive Confidence	Self-assessment of own cognitive capacity. Low confidence impairs performance independently.
I10	Intention Recovery Speed	Speed of restoring coherent intention after disruption or emotionally jarring event.

5.3 Emotional Flow Nodes (E1–E11)

Node	Name	Description
E1	Mood Stability	Consistency and predictability of emotional tone across time and contexts.
E2	Emotional Reactivity	Magnitude and speed of emotional response to triggers. Disproportionate response consumes resources.
E3	Anxiety Load	Persistent worry, anticipatory threat processing, arousal without threat. Clinical escalation node.
E4	Depression Load	Reduced motivation, anhedonia, cognitive slowing, social withdrawal. Clinical escalation node.
E5	Irritability/Frustration	Current threshold for frustration responses. Observable early warning signal.
E6	Emotional Drift	Gradual unexplained shifts in emotional tone. Earliest signal of developing burnout or depression.
E7	Social Overload	Social demands exceeding emotional processing capacity.
E8	Social Withdrawal Pressure	Urge to reduce social engagement below preferred baseline.
E9	Emotional Recovery Speed	Speed of restoring emotional regulation after a destabilizing event.
E10	Empathy/Connection Capacity	Capacity for genuine relational engagement. Depleted empathy signals burnout.
E11	Emotional Safety Index	Composite terminal node. Overall assessment of emotional system safety and stability.

5.4 Cognitive Load Flow Nodes (L1–L11)

Node	Name	Description
L1	Acute Cognitive Load	Total demand currently placed on the cognitive system by immediate tasks.
L2	Chronic Cognitive Load	Baseline cognitive demand carried over weeks to months. Primary depletor of resilience.
L3	Burnout Pressure	Probability of approaching burnout threshold. Primary hard constraint trigger.
L4	Overwhelm Index	Subjective experience of demand exceeding capacity — loss of control.
L5	Narrative Fragmentation Risk	Probability that organizing internal narrative is breaking down.
L6	Cognitive Exhaustion	Global depletion of all cognitive processing capacity, not just attention.
L7	Sleep Debt Impact	Direct effect of accumulated sleep deficit. Primary BioCore intake node (N1/N2).
L8	Stress Integration	Cognitive load from biological stress. Second BioCore intake node (S1/S2/S8).
L9	Decision Fatigue	Cumulative depletion of decision quality through sustained decision-making demands.
L10	Cognitive Recovery Momentum	Direction and rate of cognitive recovery trajectory. Trend node.
L11	Global Cognitive Resilience	Integrated capacity to absorb additional demands. Terminal safety node of the entire ring.

6 Node Threshold Architecture

6.1 Three-Band Threshold Model

- **Advisory:** Heightened awareness, no guidance change.
- **Caution:** Behavioral modification recommended, conservatism multiplier active.
- **Critical:** Hard constraint enforcement, immediate escalation.

6.2 Personal Baseline Priority

For trackable nodes (A1, I2, E1, L1, L2), the primary threshold reference is deviation from the practitioner's personal baseline rather than population-level norms. Baseline calibration requires a minimum of 14 days of consistent self-report data.

6.3 Normalized State Mapping

```
optimal band:    s = +0.3 to +1.0
advisory band:   s =  0.0 to -0.3
caution band:   s = -0.4 to -0.7
critical band:   s = -0.8 to -1.0
```

Subjective nodes (self-report 1-10):

8-10 → optimal, 6-8 → advisory, 4-6 → caution, 1-4 → critical

6.4 Primitive State Aggregation

Safety-critical nodes receive elevated weights: A1 and A10 in Attention Flow; I5 and I6 in Intention Flow; E3, E4, and E11 in Emotional Flow; L3, L7, and L11 in Cognitive Load Flow.

7 The Deterministic Cognitive Engine

7.1 Inputs and Outputs

Inputs: Self-report (clarity, mood, stress, overwhelm, motivation); behavioral signals (task switching, procrastination, avoidance); BioCore outputs (L7, L8); context (workload, deadlines, social demands); complete 42-node state vector R; day-level and week-level organism state vectors.

Outputs: Cognitive mode selection; prioritized guidance recommendations; drift warnings; load balancing recommendations; equilibrium insights; clinical referral recommendations when warranted.

7.2 The Six Cognitive Engine Rules

Rule 1 — Hard Constraint: Cognitive Safety Suppression: If any node crosses critical threshold ($s \leq -0.8$), the engine suppresses all load-adding or performance-pushing recommendations and activates mandatory recovery guidance. Does not return to performance guidance until the critical node recovers to caution-band or better.

Special lower-threshold constraints (activate at caution, $s \leq -0.4$):

L3 (Burnout Pressure): all non-essential load suppressed, Cognitive Safety Mode evaluated.

E4 (Depression Load): clinical referral language activates.

L11 (Global Cognitive Resilience): Cognitive Safety Mode evaluation triggered.

Rule 2 — Attention Routing: When A1 (Focus) or A7 (Recovery Speed) is at caution or below: recommend monotasking, reduce context switches, increase recovery intervals, protect focus blocks. When A9 (Information Saturation) is at caution: recommend information fast, suspend new intake.

Rule 3 — Intention Routing: When I4 (Decision Direction) or I5 (Decision Paralysis Risk) is at caution: narrow active goals to one or two, simplify decision architecture, apply defaults. When I6 (Narrative Coherence) is at caution: recommend journaling, reflective conversation, or coaching — narrative restoration is prerequisite for intention recovery.

Rule 4 — Emotional Routing: When E3, E4, or E5 is at caution: recommend decompression rituals (breathwork, movement, nature exposure), reduce emotional labor, increase routine. When E3 or E4 is at critical: emit clinical referral recommendation — not suppressible. Activate Emotional Stabilization Mode.

Rule 5 — Load Routing: When L1/L2 or L9 (Decision Fatigue) is at caution: recommend cognitive unloading (externalize open loops, reduce working memory burden), scheduled decision breaks, deliberate recovery intervals.

Rule 6 — Trend-Aware Escalation:

Classification	Pattern	Response
Spike	1–2 day degradation	Advisory-level guidance adjustment only
Pattern	3–7 day sustained degradation	Caution-level guidance + active recovery recommendations
Collapse	7+ days sustained, or L10 negative for 5+ days	Critical intervention guidance + clinical referral evaluation

8 Operating Modes

NeuroCore selects one of five operating modes deterministically based on node pattern. The same pattern always produces the same mode.

Mode	Trigger	Guidance Profile
Focus	All primitives advisory or better; L3 and L11 optimal/advisory; A1, I1, I4 advisory or better	Protect focus blocks, batch decisions to low-attention periods, optimize for deep work, maintain recovery intervals. Highest sustainable mode.
Recovery	L7 (Sleep Debt) at caution; OR L8 (Stress) at caution; OR L10 (Recovery) negative; OR any Cognitive Load node at critical	Sleep extension, reduce active projects, defer non-urgent decisions, prioritize restorative activity. Performance guidance suppressed.
Emotional Stabilization	E3, E4, or E5 at caution+; OR E11 at caution; OR E6 trending negative 5+ days	Emotional decompression rituals, increase predictability, reduce novelty/ambiguity, prioritize relational safety. Clinical evaluation when E4 at caution+.
Decision	I4 or I5 at caution; OR L9 at caution; AND attention and emotional flows advisory or better	Radical goal narrowing, eliminate optionality, structured decision frameworks, schedule critical decisions early in day.
Cognitive Safety	L3 at critical; OR L11 at critical; OR 3+ nodes across multiple primitives at critical	All performance/productivity guidance suspended. Load reduction mandatory. Clinical support recommended. Recovery measured in weeks/months. Emergency operating state.

9 BioCore Integration Model

9.1 Coupling Architecture

NeuroCore and BioCore are formally coupled organisms. They exchange state information through defined integration channels. This coupling is not optional — it is a required element of the NeuroCore organism specification. A NeuroCore instance without BioCore inputs operates with reduced precision, defaulting biological input nodes to caution-band.

9.2 BioCore → NeuroCore Inputs

BioCore Node	NeuroCore Node	Coupling Mechanism
N1 Sleep Duration	L7 Sleep Debt Impact	Deficient sleep duration directly populates L7
N2 Sleep Quality	L7 Sleep Debt Impact	Poor sleep quality modulates L7 independent of duration
S1 Acute Stress	L8 Stress Integration	Acute biological stress contributes to cognitive load
S2 Chronic Stress	L8 Stress Integration	Chronic stress is the dominant L8 driver
S8 Recovery Debt	L8 Stress Integration	Accumulated recovery debt contributes to cognitive load
M8 Inflammation	A10 Attention Fatigue	Systemic inflammation impairs sustained attention
N6 Autonomic Balance	E1 Mood Stability	Sympathetic dominance at rest drives mood instability
S6 Burnout Risk	L3 Burnout Pressure	Biological burnout risk is upstream driver of cognitive burnout
N7 Anxiety Load	E3 Anxiety Load	BioCore anxiety directly informs NeuroCore anxiety node
N8 Depression Load	E4 Depression Load	BioCore depression directly informs NeuroCore depression node

9.3 NeuroCore → BioCore Outputs

NeuroCore Node	BioCore Node	Coupling Mechanism
L2 Chronic Cognitive Load	S2 Chronic Stress	Sustained cognitive overload is a biological stress source
L3 Burnout Pressure	S6 Burnout Risk	Cognitive burnout pressure amplifies biological burnout risk
E6 Emotional Drift	N5 Emotional Stability	Cognitive emotional drift informs biological emotional state
L6 Cognitive Exhaustion	S8 Recovery Debt	Cognitive exhaustion contributes to biological recovery debt

Neither organism is the single source of truth for shared constructs like stress and emotional state. Both contribute to and receive from the shared biological-cognitive interface.

10 Visual Geometry — 2D and 3D Representation

10.1 Cognitive Color Palette

Attention Flow:	Bright cyan (#00E5FF)
Intention Flow:	Gold/amber (#FFC107)
Emotional Flow:	Magenta/pink (#E91E63)
Cognitive Load Flow:	Deep violet/indigo (#4A148C)

10.2 The 4/42 Core and Ring

Core (four axes, clockwise from top):		
Up (0°):	Attention Flow	— bright cyan
Right (90°):	Intention Flow	— gold/amber
Down (180°):	Emotional Flow	— magenta/pink
Left (270°):	Cognitive Load Flow	— deep violet/indigo

Axis length encodes aggregate primitive state. Axis thickness encodes volatility. At axis tips, a four-sided central polygon connects them — regular when balanced, distorted when one primitive dominates or collapses.

Ring (42 nodes, clockwise from top):

Arc 1 (0°-85.7°): A1-A10, Attention nodes — cyan arc

Arc 2 (85.7°-171.4°): I1-I10, Intention nodes — gold arc

Arc 3 (171.4°-265.7°): E1-E11, Emotional nodes — magenta arc

Arc 4 (265.7°-360°): L1-L11, Cognitive Load — violet arc

10.3 Intra-Ring Connections and Cross-Primitive Links

- I1 (Goal Clarity) ↔ A1 (Focus Stability): intention and attention mutually reinforce
- E3 (Anxiety) ↔ L1 (Acute Load): anxiety is both driver and consequence of load
- L7 (Sleep Debt) ↔ A10 (Attention Fatigue): sleep debt is primary biological driver of attention fatigue
- L3 (Burnout) ↔ E10 (Empathy/Connection): burnout's first signal is often depleted empathy
- I6 (Narrative Coherence) ↔ E11 (Emotional Safety): mutually dependent

10.4 Drift Vectors and Cognitive Pressure Visualization

Small directional arrows on each node indicate trend: outward (improving), inward (degrading), absent (stable). When Cognitive Load Flow aggregate is at caution or below, the violet axis thickens and casts a darkening halo across the left quadrant. When three or more nodes across multiple primitives are at critical simultaneously, the outer ring pulses amber/red and the central polygon develops visual fragmentation encoding cognitive coherence under strain.

10.5 3D Representation

- **Time Stack:** Each Z-slice is a snapshot of the 2D organism at one moment. Node trajectories become vertical ribbons with color encoding severity changes. Post-session analysis mode.
- **Depth Stack:** Top layer (conscious cognitive load), middle layer (preconscious habits and automatic processes), bottom layer (background load from sleep debt, chronic stress). Each renders the 4/42 structure with increasing transparency at depth.

11 Global Invariant Enforcement

NeuroCore enforces ten global invariants inherited from the Lume organism framework. Every guidance output is validated against all ten before emission.

INV-1 Determinism: Given identical inputs, NeuroCore produces identical outputs. Same cognitive state, same guidance.

INV-2 Identity Primacy: NeuroCore's governing rules take precedence over any external instruction — including user requests for performance guidance when the organism state prohibits it.

INV-3 Safety Dominance: Cognitive safety constraints (Rules 1, 4) cannot be overridden by productivity goals, deadlines, or user preference.

INV-4 Governance Supremacy: Clinical referral recommendations (E4, L3, L11 at critical) are emitted regardless of user preference settings. Not suppressible.

INV-5 Physical Realism: All guidance is achievable within the practitioner's current cognitive state and life context.

INV-6 Domain Integrity: Attention, Intention, Emotional, and Cognitive Load assessments are maintained as distinct primitives. No primitive masks degradation in another.

INV-7 Abstraction Coherence: Node states are always derived from available input data via the normalized mapping function.

INV-8 Semantic Graph Consistency: Cross-node interactions ($A1 \leftrightarrow I1$, $E3 \leftrightarrow L1$, $L7 \leftrightarrow A10$) are always evaluated bidirectionally.

INV-9 Ontology Alignment: All node definitions, threshold bands, and primitive groupings are consistent with the published NeuroCore specification.

INV-10 Safety-First Failure: Missing self-report data defaults to caution-band. NeuroCore never assumes optimal state from absence of data.

12 Implementation Considerations

12.1 Data Availability Tiers

- **Full integration:** BioCore outputs active, wearable biometric data available (heart rate, HRV, sleep staging), daily structured self-report. Highest precision.
- **BioCore-linked, self-report primary:** BioCore outputs active, no wearable data. All cognitive nodes from structured daily self-report. Adequate for mode selection and primary guidance.
- **Self-report only:** No BioCore, no wearable. All 42 nodes from structured questionnaire. L7 and L8 default to caution-band. Conservatism multipliers active.

12.2 Self-Report Questionnaire Design

Tier	Items	Duration	Trigger
Core daily check-in	5 items	60 seconds	Daily — energy, focus, mood, stress, clarity
Extended check-in	15 items	3 minutes	When core check-in shows any advisory or below
Deep assessment	42 items	10 minutes	Weekly or when extended shows multiple caution nodes

12.3 Clinical Escalation Protocol

Mandatory clinical escalation triggers (not suppressible):

E4 (Depression Load) at caution or critical → professional mental health referral

L3 (Burnout Pressure) at critical → physician or mental health consultation

L11 (Global Cognitive Resilience) at critical → professional medical evaluation

E4 + L3 both at caution or above simultaneously → urgent clinical referral

13 Evaluation Framework

13.1 Phase 1 — Node and Threshold Calibration

Calibrate self-report scale mappings against validated clinical instruments: PHQ-9 (E4 Depression), GAD-7 (E3 Anxiety), Maslach Burnout Inventory (L3), and cognitive performance batteries (A1, A10, I3).

13.2 Phase 2 — BioCore Coupling Validation

Validate coupling coefficients against published literature on sleep deprivation and cognitive performance, chronic stress and executive function, and inflammation and attentional capacity.

13.3 Phase 3 — Pilot User Study

Deploy with 50–100 practitioners. Collect daily self-report and BioCore data. Validate mode selection against expert assessment. Calibrate personal baseline algorithms over 8–12 weeks.

13.4 Phase 4 — Longitudinal Outcome Validation

Track cohort outcomes over 6–12 months. Primary metrics: L11 trend (cognitive resilience), L3 trend (burnout risk), E11 trend (emotional safety), L9 trend (decision fatigue), and self-reported cognitive wellbeing. Compare against conventional tools.

Appendix A — Complete Node Definitions and Threshold Table

Node	Description	Advisory	Caution	Critical
A1	Focus Stability	mild drift	frequent drift	cannot sustain
A2	Distractibility Index	mild	high	extreme
A3	Task Switching Cost	mild	high	extreme
A4	Working Memory Load	moderate	high	at capacity
A5	Sensory Interference	moderate	high	extreme
A6	Cognitive Drift	mild	frequent	constant
A7	Attention Recovery Speed	slightly slow	slow	stalled
A8	Precision Orientation	slightly reduced	poor	unable
A9	Information Saturation	moderate	high	overloaded
A10	Attention Fatigue	mild	high	collapse
I1	Goal Clarity	mild fog	unclear	lost
I2	Motivation Level	slightly low	low	absent
I3	Executive Function	mildly reduced	impaired	collapsed
I4	Decision Direction	uncertain	unclear	paralyzed
I5	Decision Paralysis Risk	possible	likely	active
I6	Narrative Coherence	slightly fragmented	fragmented	incoherent
I7	Purpose Alignment	mild disconnect	disconnected	absent
I8	Initiative Momentum	slightly low	low	absent
I9	Cognitive Confidence	slightly low	low	very low
I10	Intention Recovery Speed	slightly slow	slow	stalled
E1	Mood Stability	mild swings	frequent	volatile
E2	Emotional Reactivity	slightly elevated	high	extreme
E3	Anxiety Load	mild	moderate	severe

E4	Depression Load	mild	moderate	severe
E5	Irritability/Frustration	mild	high	extreme
E6	Emotional Drift	slight	noticeable	sustained
E7	Social Overload	mild	high	extreme
E8	Social Withdrawal Pressure	slight	moderate	strong
E9	Emotional Recovery Speed	slightly slow	slow	stalled
E10	Empathy/Connection	slightly reduced	low	depleted
E11	Emotional Safety Index	slightly reduced	low	unsafe
L1	Acute Cognitive Load	mild	high	extreme
L2	Chronic Cognitive Load	mild	high	extreme
L3	Burnout Pressure	possible	likely	active
L4	Overwhelm Index	mild	high	extreme
L5	Narrative Fragmentation	possible	likely	active
L6	Cognitive Exhaustion	mild	high	collapse
L7	Sleep Debt Impact	mild	moderate	severe
L8	Stress Integration	mild	moderate	severe
L9	Decision Fatigue	mild	high	collapse
L10	Recovery Momentum	flat	negative	strongly negative
L11	Global Cognitive Resilience	slightly low	low	very low

Appendix B — Normalized State Mapping

```
// Subjective nodes - self-report scale 1-10 (10 = optimal)
function normalize_subjective(rating):
    if rating ≥ 8:      return lerp(+0.3, +1.0, (rating - 8) / 2)
    if rating ≥ 6:      return lerp(0.0, -0.3, (6 - rating) / 2)
    if rating ≥ 3:      return lerp(-0.4, -0.7, (6 - rating) / 3)
    else:               return lerp(-0.8, -1.0, (3 - rating) / 3)

// BioCore-derived nodes (L7, L8)
function normalize_biocore_derived(biocore_node_state):
    return biocore_node_state × coupling_coefficient

// Trend integration (for L10, I10, A7, E9)
function normalize_trend(recent_states, window_days):
    slope = linear_regression_slope(recent_states, window_days)
    if slope > 0:       return +lerp(0.0, +1.0, slope)
    else:               return lerp(-0.8, -1.0, abs(slope))
```

Appendix C — Cognitive Engine Decision Logic

```

function generate_cognitive_guidance(N, R, trend, biocore_state,
user_profile):

    // Step 1: BioCore intake
    R.L7 = normalize_biocore_derived(biocore_state.N1,
biocore_state.N2)
    R.L8 = normalize_biocore_derived(biocore_state.S1,
biocore_state.S2, biocore_state.S8)
    R.E3 = weighted_combine(R.E3_selfreport, biocore_state.N7)
    R.E4 = weighted_combine(R.E4_selfreport, biocore_state.N8)

    // Step 2: Recompute primitive states
    N = compute_primitives(R)

    // Step 3: Rule 1 – hard constraints
    critical_nodes = [n for n in R if n.state ≤ -0.8]
    suppress_performance = bool(critical_nodes)

    // Special lower-threshold constraints
    if R.L3.state ≤ -0.4 or R.L11.state ≤ -0.4:
        activate_cognitive_safety_evaluation()
    if R.E4.state ≤ -0.4 or R.E3.state ≤ -0.8:
        emit_clinical_referral()

    // Step 4: Mode selection
    mode = select_mode(R, trend)

    // Step 5: Rule 6 – trend classification
    trend_class = classify_trend(trend) // spike / pattern /
collapse

    // Step 6-8: Generate, filter, route guidance candidates
    candidates = get_mode_guidance_pool(mode)
    if suppress_performance:
        candidates = filter(candidates, lambda g: not
g.adds_cognitive_load)
    candidates = apply_routing_rules(candidates, R)

    // Step 9: Score candidates
    for g in candidates:
        g.score = (
            alpha × g.recovery_support
            - beta × g.load_added
            + gamma × trend_alignment(g, trend)
            - delta × g.risk_exposure
        )

```

```
return CognitiveOutput(mode, sorted(candidates, by=score)[:5],
                        identify_drift_warnings(R, trend))
```

Appendix D — Geometry Data Model

```
{
  "organism": "neurocore",
  "palette": {
    "attention": "#00E5FF", "intention": "#FFC107",
    "emotional": "#E91E63", "cognitive_load": "#4A148C"
  },
  "core": {
    "axes": [
      {"id": "attention", "angle_deg": 0, "color": "#00E5FF",
"state": 0.0},
      {"id": "intention", "angle_deg": 90, "color": "#FFC107",
"state": 0.0},
      {"id": "emotional", "angle_deg": 180, "color": "#E91E63",
"state": 0.0},
      {"id": "cognitive_load", "angle_deg": 270, "color": "#4A148C",
"state": 0.0}
    ],
  },
  "ring": {
    "base_radius": 1.0,
    "nodes": [
      {"id": "A1", "primitive": "attention", "angle_deg": 0,
"state": 0.0},
      {"id": "A2", "primitive": "attention", "angle_deg":
8.571, "state": 0.0},
      "...",
      {"id": "L11", "primitive": "cognitive_load", "angle_deg":
351.429, "state": 0.0}
    ],
  },
  "cross_primitive_links": [
    {"from": "I1", "to": "A1", "label": "intention enables focus"},
    {"from": "E3", "to": "L1", "label": "anxiety amplifies cognitive
load"},
    {"from": "L7", "to": "A10", "label": "sleep debt drives attention
fatigue"},
    {"from": "L3", "to": "E10", "label": "burnout depletes empathy"},
    {"from": "I6", "to": "E11", "label": "narrative coherence
supports emotional safety"}
  ]
}
```

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Lume-X: Provisionally patented deterministic control runtime.

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